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Saveetha Institute of Medical And Technical Sciences

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Capstone Project Proposed Title:

VMware Virtualization-Based Cloud Computing for Weather Forecasting and Flood Risk Prediction Using Big Data Analytics and Virtual Machines

Team no. 5

CSA15 – Cloud computing and Big Data analytics

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Abstract

- Flooding can be associated with intense rainfall, river conditions and local catchment characteristics.
- The proposed system brings together weather forecasts and historical rainfall datasets for analysis.
- VMware virtual machines provide an isolated environment for the operating system and project tools.
- Cloud-ready storage and big data processing support handling of larger weather records.
- Rainfall indicators are compared with documented thresholds to estimate a risk category.
- A dashboard presents location, data timestamp, forecast information and configurable alerts.

Aim and Objective

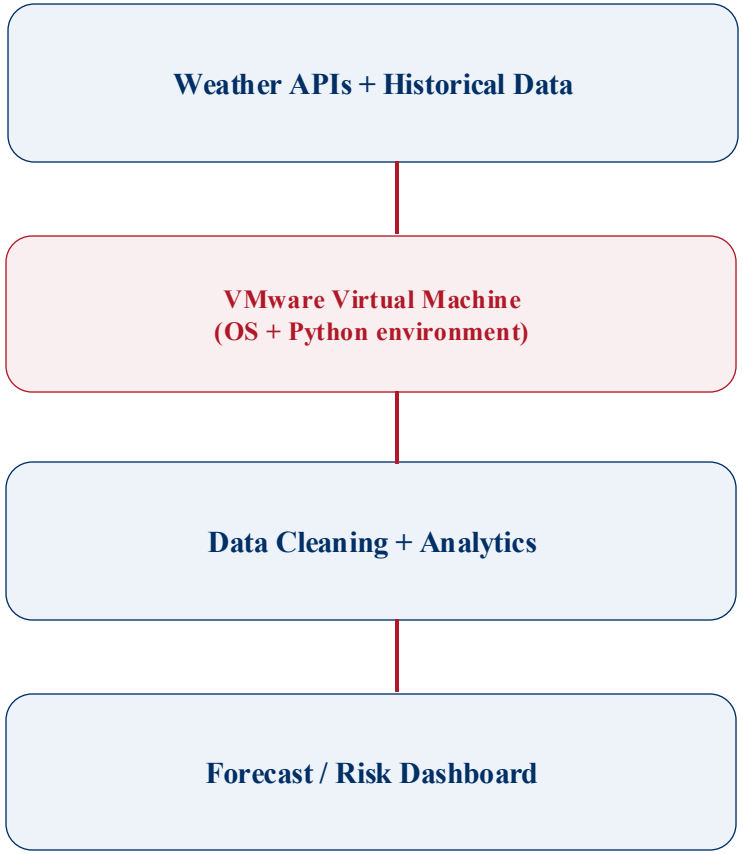
AIM: Develop a VMware-virtualized cloud-computing workflow for weather analysis and flood-risk awareness.

OBJECTIVES

- Collect weather forecasts and historical rainfall records from documented sources.
- Prepare a virtual machine environment for running project components.
- Clean and process data; use Apache Spark if required by the dataset size and available resources.
- Calculate rainfall indicators and apply transparent risk-classification rules.
- Visualize weather and location-wise risk status in a dashboard.
- Test the workflow with different conditions and document its limitations.

Introduction

- Flood hazards can affect communities, infrastructure and transport.
- Weather data may come from APIs, historical archives and local measurements.
- Large historical datasets need repeatable cleaning and analysis.
- Virtual machines provide a controlled environment for software deployment.
- Cloud computing supports flexible storage, processing and access.



Literature Review & Requirement Analysis

S.No	Year of Publication	Author	Title of the Journal / book	Methodology used	Algorithm used	Key Findings
1	2022	Zhang, Ragettli, Molnar, Fink & Peleg	Generalization of an Encoder-Decoder LSTM model for flood prediction in ungauged catchments. Journal of Hydrology.	Local and regional deep-learning models tested for catchments with limited gauge data.	Encoder–Decoder LSTM (ED-LSTM).	Outperformed a conceptual hydrological model and showed promise where stream-gauge data are sparse.
2	2022	Cao et al.	Multi-step-ahead flood forecasting using an improved BiLSTM-S2S model. Journal of Flood Risk Management.	Compared multi-step runoff forecasts against LSTM-Seq2Seq and MLP baselines.	BiLSTM-Seq2Seq.	Improved 6-hour-ahead runoff metrics and reduced prediction time lag versus the comparison models.
3	2023	Zou, Wang, Lei & Li	A novel multi-step ahead forecasting model for flood based on time residual LSTM. Journal of Hydrology.	Probabilistic, multi-step flood forecasting with time-series models.	Residual LSTM (ResLSTM) and DeepAR-based models.	Highlighted probabilistic forecasts as useful for representing uncertainty in flood prediction.
4	2023	Zhou, Chen & Liu	Application of a New Hybrid Deep Learning Model That Considers Temporal and Feature Dependencies in Rainfall–Runoff Simulation. Remote Sensing.	Hourly streamflow forecasting using a hybrid model to capture temporal and feature relationships.	SA-CNN-LSTM (self-attention, CNN and LSTM).	Combines attention and deep-learning components to model rainfall–runoff dependencies.
5	2023	Khatun, Chatterjee, Sahu & Sahoo	A novel smoothing-based long short-term memory framework for short-to medium-range flood forecasting. Hydrological Sciences Journal.	Tested forecasts up to five days ahead in India's Mahanadi River basin against benchmark models.	Smooth-LSTM; compared with LSTM, ANN and MIKE11 NAM-HD.	Explored improved short-to medium-range forecasting for a monsoon-influenced basin.

Literature Review & Requirement Analysis

FUNCTIONAL REQUIREMENTS:

- Collect and validate data
- Process rainfall indicators
- Classify risk and show alerts

NON-FUNCTIONAL REQUIREMENTS:

- Reliability and usability
- Scalability and maintainability
- Cost awareness and traceability

Problem Statement

Weather and rainfall information can be spread across sources and may differ in format, timing and location coverage.

- Large historical records can be time-consuming to clean and compare manually.
- Risk interpretation requires clear thresholds or a validated prediction method.
- VM and processing environments need a repeatable configuration.
- Users need location-aware visualizations and understandable alerts.
- Missing or stale readings must be flagged to avoid misleading risk labels.

Existing System

COMMON EXISTING APPROACHES

- Weather portals and forecast services provide source-specific information.
- Historical datasets are often analyzed separately for a particular study.
- Fixed thresholds are easy to interpret but require local calibration.
- Forecasting models need appropriate data and performance evaluation.

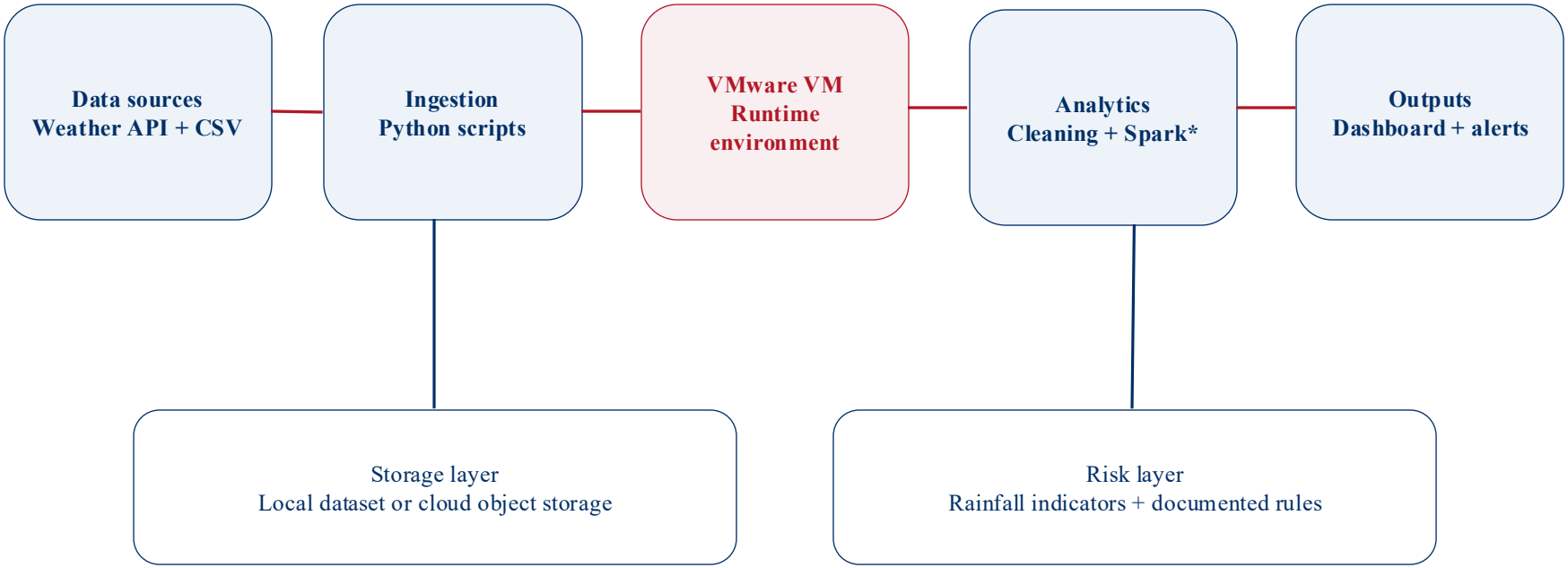
PROJECT GAP / CONTRIBUTION

- Combine available data in one workflow.
- Run project software in a VMware virtual machine.
- Keep preprocessing and risk rules repeatable.
- Show data source, timestamp, category and limitations together.

Proposed System

1. Data collection	Weather API + historical rainfall records
2. Virtualized environment	VMware VM with OS, Python and dependencies
3. Data preparation	Validate timestamps, units, missing and duplicate values
4. Big data analytics	Process records; Spark can be used when appropriate
5. Risk estimation	Accumulated rainfall + configured thresholds / validated model
6. Dashboard and alerts	Forecast, location, risk level, timestamp and alert history

Overall System Architecturel Design



Data flow: collect → validate → store/process → estimate risk → visualize

Sustainable Development Goal (SDG) Implementation

SDG 11
Sustainable Cities and Communities

SDG 13
Climate Action

- Improve access to understandable weather and risk information.
- Support awareness and preparedness planning.
- Encourage data-informed discussion of local conditions.
- These are intended contributions; the prototype alone does not prove reduced flood losses.

Proposed Methodology and Algorithm

1. Collect	Fetch forecast and historical rainfall records
2. Validate	Check units, time, missing and stale data
3. Aggregate	Calculate rolling / accumulated rainfall
4. Compare	Compare with locally configured thresholds
5. Classify	Assign risk category only when inputs suffice
6. Display	Store result and update dashboard / alerts

Design & Module Planning

MODULE 1 — DATA COLLECTION

Weather API integration; historical dataset import; source and timestamp tracking

MODULE 2 — VM + STORAGE + PROCESSING

VMware VM setup; cloud-ready storage; cleaning; Spark processing if implemented.

MODULE 3 — RISK + ALERTS

Threshold detection; risk classification; configurable alerts; alert history.

MODULE 4 — DASHBOARD

Streamlit interface; Plotly charts; location-wise display; historical trends.

Conclusion

- The proposed architecture connects weather data, a VMware virtual environment and rainfall analysis.
- Repeatable preprocessing can make historical data easier to examine.
- Threshold-based classification is interpretable, but thresholds require local justification.
- The dashboard can show location, source, timestamp, risk category and configurable alerts.
- Evaluation with historical events is needed before making claims about predictive accuracy.

References

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- [8] United Nations. Sustainable Development Goals. https://sdgs.un.org/goals